Topic 8: Energetics I

In order to develop their practical skills, students should be encouraged to carry out a range of practical experiments related to this topic. Possible experiments include a wide variety of calorimetry experiments involving displacement and neutralisation reactions, investigating the enthalpy of combustion of a homologous series of alcohols.

Mathematical skills that could be developed in this topic include plotting and extrapolating graphs of temperature rise against time for displacement reactions, calculating enthalpy changes in J and in kJ mol^{-1}, using algebra to solve Hess's law problems, calculating enthalpy changes using bond enthalpies.

Within this topic, students can consider how the use of Hess's Law can facilitate the study of energy changes in reactions which are not directly measureable. They can also consider the value of a general chemical concept, such as mean bond enthalpy, and why the use of a simplification such as this has some benefits, as well as some short-comings.

Students should:
1. know that standard conditions are 100 kPa and a specified temperature, usually 298 K
2. know that the enthalpy change is the heat energy change measured at constant pressure
3. be able to construct and interpret enthalpy level diagrams showing an enthalpy change, including appropriate signs for exothermic and endothermic reactions *Activation energy is not shown in enthalpy level diagrams but it is shown in reaction profile diagrams.*
4. be able to define standard enthalpy change of: i reaction ii formation iii combustion iv neutralisation
5. understand experiments to measure enthalpy changes in terms of: i processing results using the expression: energy transferred = mass x specific heat capacity × temperature change ($Q=mc\Delta T$) ii evaluating sources of error and assumptions made in the experiments *Students will need to consider experiments where*: ○ *substances are mixed in an insulated container and the temperature change is measured* ○ *enthalpy of combustion is measured, such as using a series of alcohols in a spirit burner* ○ *the enthalpy change cannot be measured directly*.

Students should:
7. be able to construct enthalpy cycles using Hess’s Law
8. be able to calculate enthalpy changes from data using Hess’s Law
CORE PRACTICAL 8: To determine the enthalpy change of a reaction using Hess’s Law
9. know what is meant by the terms ‘bond enthalpy’ and ‘mean bond enthalpy’
10. be able to calculate an enthalpy change of reaction using mean bond enthalpies and explain the limitations of this method of calculation
11. be able to calculate mean bond enthalpies from enthalpy changes of reaction